# SATYAJIT S

B.Tech. Computer Science • IIITDM Kancheepuram
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Computer science undergraduate working on physical AI: learned policies for robotic manipulation and control. Vision-language-action models, imitation learning, diffusion models and reinforcement learning, deployed on real manipulators and mobile robots.

## EDUCATION

**IIITDM Kancheepuram** – B.Tech. Computer Science *2024 – 2028 | CGPA: 8.90*
**The Lords International School, Chennai** – Cambridge Board (A-Levels) *2023 – 2024 | 80.8%*

## RESEARCH EXPERIENCE

**e-Yantra Robotics Research Intern** | IIT Bombay *May 2026 – Jul 2026*

- Trained and benchmarked four Vision-Language-Action (VLA) policies – ACT (Action Chunking with Transformers), SmolVLA, $\pi_0$ and GR00T – on SO-101 bimanual arms using the LeRobot framework, comparing task success across architectures.
- Collected expert demonstration datasets for imitation learning and deployed the policies on bimanual stacking, precision pick-and-place, and arranging markers and cups.
- Studied the VLA stack component by component – vision encoder, VLM backbone, action decoder, variational encoder – to diagnose failure modes rather than treat policies as black boxes.
- Selected for the eYSIP research programme on the strength of a top-20 national finish in the e-Yantra Robotics Competition.

**RL-Based QA Tester for Games** | ViZDoom, PyTorch, PPO, Go-Explore *Oct 2026 – Present*

- Built the environment end to end: a declarative map specification compiles to a playable UDMF level plus ground-truth JSON, driving a ViZDoom simulation where six game-agnostic oracle families detect injected defects.
- Training the RL tester now: PPO with count-based curiosity and Go-Explore archive resets, benchmarked against a task-reward-only baseline and scored on precision and recall.

## PROJECTS

**TD3-based Autonav** | Reinforcement Learning Navigation *Sep 2026 – Oct 2026*

- Developed a TD3 agent for TurtleBot3 maze navigation in Gazebo (ROS 2 Humble, Stable-Baselines3) using LiDAR and relative-goal observations, with BFS geodesic reward shaping.
- Trained on a single maze and transferred to unseen layouts without retraining, reaching ∼**98% success** on the training map.

**Grayscale2RGB** | Diffusion Models for Image Colourization *Jun 2026 – Jul 2026*

- Implemented a conditional Denoising Diffusion Probabilistic Model (DDPM) from scratch, with a custom UNet and spatial self-attention, using Classifier-Free Guidance at inference.

**Autonomous Holonomic Bot** | e-Yantra Robotics Competition (IIT Bombay) *Sep 2025 – May 2026*

- Placed **top 20 of 300+ teams nationwide** over six months of national software and hardware rounds, building a holonomic robot with a PID-controlled omnidirectional drive and an ArUco vision pipeline for real-time pose estimation and autonomous pick-and-place.

## ROBOTICS EXPERIENCE

**MaRS Student Rover Society (IIITDM)** | Software Team Member *Jan 2025 – Present*

- Developed a Cartesian control pipeline for a custom-built 5-DOF manipulator using MoveIt 2: the operator commands the end-effector directly while inverse kinematics resolves joint angles.
- Contributed to the rover software stack that placed the team **10th at the International Rover Challenge (IRC)**.

## TECHNICAL SKILLS

- **Learning:** PyTorch, TensorFlow, Stable-Baselines3, LeRobot, imitation learning, policy-gradient RL (PPO, TD3, GRPO), diffusion models, vision-language-action policies
- **Robotics & Control:** ROS 2, MoveIt 2, Nav2, Gazebo, MuJoCo, RViz 2, inverse kinematics, PID control, teleoperation, LiDAR SLAM (slam_toolbox), ArUco pose estimation
- **Vision:** OpenCV, YOLO, Scikit-learn • **Languages:** C, C++, Python, JavaScript • **Tools:** Git, Linux, Docker
- **Certifications:** Reinforcement Learning (NPTEL) | Udemy: Self-Driving – Map & Localization | Self-Driving – Odometry and Control